

NETFLIX CONTENT ANALYSIS DASHBOARD

Project Overview: Netflix Content Catalog Analysis

Built with Power BI | Author: Yatin Kondal

1. What this project is

This project is an interactive Power BI dashboard that shows what is inside the Netflix content catalog. It takes a public list of 8,807 Netflix titles (movies and TV shows) and turns it into one screen where a reader can see the mix of content, the most common genres, how long titles run, and which directors appear most often.

The project covers the full workflow: cleaning the raw data, building a data model, writing DAX measures, and designing an interactive report on top of them.

2. Why it was built

The scenario is a Netflix Content Strategy team that needs a quick, visual way to understand its catalog before making acquisition decisions. The dashboard is meant to answer questions such as:

- How is the catalog split between Movies and TV Shows?
- Which genres have the most titles?
- How long is a typical movie, and how many seasons does a typical TV show run?
- Which directors have the most titles with a rating on record?
- How do these answers change for one country, one genre or one director?

3. The data

Source file: netflix_titles.csv, one row per title, 12 columns.

Column	What it holds
show_id	Unique ID for each title (s1, s2, ...)
type	Movie or TV Show
title, description	Name and a one-line summary
director, cast	People credited; often several names in one cell, and director is blank for 2,634 titles
country	Production country or countries, comma separated; blank for 831 titles
date_added, release_year	When Netflix added the title; year it was released (1925 to 2021)
rating	Maturity rating such as TV-MA, TV-14, PG-13 (4 blanks)
duration	Minutes for movies ("90 min"), seasons for TV shows ("2 Seasons")
listed_in	Genres, comma separated (for example "Dramas, International Movies")

Catalog at a glance. 6,131 movies (69.6%) and 2,676 TV shows (30.4%), 8,807 titles in total.

4. How the data was prepared

The raw file is not ready for charting, because several columns pack many values into one cell. The Power Query steps in the report fix this:

- **Missing values.** Blank directors became "Director Missing" and blank countries became "United States" (the most frequent country).
- **Dates and numbers.** date_added was converted to a real date and release_year to a whole number.
- **Duration split.** duration was split into a number and a unit, then into two columns: "movie duration" (minutes, movies only) and "TV Duration" (seasons, TV shows only).
- **Primary category.** The first genre in listed_in was pulled into its own column so each title belongs to one main genre.
- **Cleaning.** Text in country, director and category was trimmed and cleaned of stray characters.
- **Lookup tables.** Countries, Directors and category_mapping were created by splitting multi-value cells into one row per value.

5. Data model and measures

The model uses one main fact table, netflix_titles, with three helper tables linked to it by show_id (many-to-one). Because of these links, picking a country, director or genre in a slicer filters every visual on the page.

Table	Role
netflix_titles	Main table with one row per title
Countries	One row per title and country; feeds the Country slicer
Directors	One row per title and director; feeds the Director slicer
category_mapping	One row per title and genre, used to analyse genres

Five DAX measures drive the cards and charts:

Measure	Definition
Total Titles	COUNTROWS(netflix_titles)
Total Movies	CALCULATE([Total Titles], netflix_titles[type] = "movie")
Total TV Shows	CALCULATE([Total Titles], netflix_titles[type] = "tv show")
Avg Movie (mins)	AVERAGE(netflix_titles[movie duration])
Avg TV Duration (seasons)	AVERAGE(netflix_titles[TV Duration])

6. The dashboard

The report is one 1920 x 1080 page with a dark-red, Netflix-style theme. Reading from top to bottom:

Area	Visual	What it shows
Top banner	Title text box	"NETFLIX Content Analysis Dashboard"

Area	Visual	What it shows
Row of 5 KPI cards	Cards	Avg TV duration (seasons), Avg movie length (mins), Total Titles, Total TV Shows and Total Movies
Left, upper	Pie chart	Movie vs TV Show share of titles, with counts and percentages
Left, lower	Bar chart	Seasons per TV show title, sorted from longest down
Centre	Bar chart	Titles per primary category, sorted highest first
Right, upper	Three dropdown slicers	Primary Category, Country, Director
Right, lower	Bar chart	Count of rating per director, highest first, with "Director Missing" excluded

The design follows Netflix branding: red visuals on a dark background, with white text.

7. Interactivity

Visuals cross-filter one another: clicking a slice, a bar or a slicer value re-computes the rest of the page. Three useful ways to explore it, and the behaviour to expect, are:

- **Click "Movie" in the pie chart.** The category chart shows movie genres only, and the TV-season chart has nothing to show because movies have no seasons.
- **Click "TV Show" in the pie chart.** The category and duration visuals switch to TV genres and season counts.
- **Set the Country slicer to France.** All cards and charts recalculate for the 393 titles that list France among their countries.

8. Key findings

These numbers come from the full, unfiltered catalog.

- **Movies dominate.** Almost 7 in 10 titles are movies.
- **Dramas lead.** Dramas is the biggest primary category with 1,600 titles, followed by Comedies (1,210), Action & Adventure (859), Documentaries (829) and International TV Shows (774).
- **Movies are about 100 minutes long.** The average is 99.6 minutes; the middle value is 98 minutes, and lengths range from 3 to 312 minutes.
- **Most TV shows are short.** The average is 1.76 seasons, and 67% have only one season.
- **Genre tags.** Counting every genre tag on every title (42 distinct genres), International Movies (2,752), Dramas (2,427), Comedies (1,674), International TV Shows (1,351) and Documentaries (869) are the five biggest.
- **Prolific directors.** Rajiv Chilaka has the most titles with a rating (19), followed by the team of Raúl Campos and Jan Suter (18), then Marcus Raboy and Suhas Kadav (16 each).

Quick answers from the data

Question	Answer
Which TV show has the highest runtime?	Grey's Anatomy, with 17 seasons (NCIS and Supernatural follow with 15)
What is the runtime of Dramas directed by Christophe Charrier?	"I Am Jonas" (Dramas, Independent Movies, International Movies) runs 82 minutes

9. Notes and limits

- The director chart shows the top directors only. Director names are counted as written in the file. Titles with two directors, such as Raúl Campos and Jan Suter, appear as one combined bar.
- The Country slicer lists every country named in a title, so a co-produced title appears under each of its countries and the country counts add up to more than the number of titles.
- Missing countries were filled with "United States", which may overstate US titles.
- Three movies have no duration, so the movie average uses 6,128 of 6,131 movies.
- The data stops at titles released in 2021 and says nothing about viewing numbers, so "popular" here means "most titles", not "most watched".

10. Tools

Power BI Desktop (Power Query, data model, DAX, report visuals); source data netflix_titles.csv.
Report file: Netflix.pbix.